

A Study: Text Translation and Transliteration for Ottoman Tax Records

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Abstract—Ottoman Turkish, written in a historical Arabic-based script, has challenges for machine translation due to its complex orthography, limited parallel corpora, and degradation in historical documents. This study addresses the translation and transliteration of handwritten Ottoman tax records (*Temettuat* Notebooks), aiming to support historical research and language preservation. We explore two approaches: multimodal language models, and an OCR pipeline using the TrOCR architecture fine-tuned on annotated data. Despite recent advancements in MLLMs, these models performed poorly due to domain mismatches and lack of exposure to Arabic-script handwriting. The OCR-based approach, although it is limited by data size and handwriting variability, produced slightly more structured outputs. Our findings highlight the need for domain-adapted models and larger datasets for low-resource historical scripts, contributing to the broader field of history, specifically Ottoman studies.

Index Terms—Machine Translation; Ottoman Turkish; Multimodal LLM; OCR; Handwritten Text Recognition

I. INTRODUCTION

Machine translation has become essential in breaking down language barriers, and enabling the global exchange of information, literature, and culture. Even though considerable progress has been made in translating high-resource languages, low-resource and historically significant languages like Ottoman Turkish remain underexplored. Ottoman Turkish, used in the administrative, literary, and scholarly works of the Ottoman Empire for over six centuries, holds historical, linguistic, and cultural value. Current translation efforts have been boosted by advancements in neural machine translation (NMT), dataset development, and fine-tuning of large language models (LLMs). However, due to its complex script, orthographic variations, and limited parallel corpora, translating Ottoman Turkish still has unique challenges. Additionally, the limited of digitized and annotated parallel corpora increases the problem and limits the applicability of machine translation methods.

This work studies the need for linguistic preservation and accessibility by developing a machine translation pipeline capable of converting handwritten Ottoman tax records which are known as *Temettuat* Notebooks

into structured transliterations or translations in Latin-script Turkish or English. These records document socio-economic and demographic data in a household level. They hold critical importance to historians and researchers in the fields of Ottoman studies, historical sociology, and digital humanities. However, the documents have challenges due to the handwritten, non-standardized, and often degraded nature of the archival pages.

To explore these challenges, our research studies two approaches multimodal large language models (MLLMs) and traditional OCR-based techniques. The first approach utilizes the recent advancements in multimodal AI, which integrates vision and language understanding. These models are typically trained on vast datasets of image-text pairs and have shown promise in tasks like image captioning, visual question answering, and grounded dialogue. Our hypothesis was that such models, if adapted properly, could learn to associate specific visual regions in historical document images with their corresponding textual labels, in this case their translations.

Despite the theoretical promise of MLLMs, we faced the expected limitations. Multimodal models are often optimized for contemporary scenes and textual content in Latin scripts commonly in printed format, making them not suited for recognizing and interpreting handwritten Arabic-script documents like the *Temettuat* records. Domain mismatch, limited training data, and the lack of exposure to historical scripts led to inconsistent and semantically unrelated outputs.

Given these challenges, we shifted our focus to a more conventional and narrow method and worked on an OCR-based pipeline using Transformer-based vision-language models specifically designed for text recognition, such as TrOCR. By constructing a dataset with Ottoman handwritten tokens annotated with Latin transliterations, we fine-tuned TrOCR to perform isolated word recognition. Although the performance was slightly more successful, as it identified couple initial characters or numeric figures and surpassed the MLLM-based approach in producing structured and partially accurate outputs, it still did not produce great results.

This exploration not only highlights the current limitations of more general multimodal models in low-resource,

historical document domains but also shows the importance of specialized OCR systems in practical digital humanities workflows. More broadly, this work contributes to the fields of low-resource machine translation, historical document processing, and multimodal model adaptation. It provides a foundation for future developments in computational Ottoman studies and the design of domain-aware models for morphologically rich and underrepresented languages.

The primary objective of this study was to develop a machine translation pipeline capable of converting handwritten Ottoman tax records known as *Temettuat* Notebooks into structured transliterations or translations in Latin-script Turkish or English. These documents are highly valuable to historians due to the socio-economic and demographic data they contain, but has significant challenges due to their handwritten Arabic script form.

II. RELATED WORK

Research on translating Ottoman Turkish to English remains limited due to the language’s complex script variations and lack of parallel corpora. In this section we review developments in machine translation for Ottoman Turkish, available datasets, and language models for low-resource languages.

A. Machine Translation for Ottoman Turkish

Early Ottoman Turkish translation systems relied on statistical machine translation (SMT), but recent efforts have shifted toward neural machine translation (NMT). The study in [1] developed an NMT system using sentence alignment techniques (SentAlign, VecAlign, LASER, and LaBSE) to construct parallel corpora. The authors fine-tuned Opus-MT, GPT-3.5 Turbo, GPT-4 Turbo, Gemini, and Cohere Aya, showing that large language models (LLMs) exhibit emergent translation capabilities despite limited training data.

Beyond direct machine translation, transliteration plays a crucial role in Ottoman Turkish processing. The study in [2] introduced OtoL, a rule-based transliteration system that converts Ottoman script into Latin-based Turkish. This approach aids in preprocessing text before translation, addressing character mapping and orthographic complexities. Another work [3] explored end-to-end Ottoman-to-Turkish conversion, integrating OCR, alphabet transliteration, and machine translation in a single pipeline, highlighting the importance of multi-step processing.

Another recent effort [4] assessed the translation capabilities of Google Gemini 1.5 Pro on 18th-century Ottoman Turkish manuscripts, revealing challenges in handling historical and sensitive content due to AI safety mechanisms. Moreover, research in [5] demonstrated how NLP techniques (NER, topic modeling, and sentiment analysis) could aid in processing Ottoman Turkish texts, particularly in historical research such as Evliya Çelebi’s travelogue.

OCR plays a crucial role in digitizing Ottoman texts, especially handwritten or printed archival materials. A significant advancement is the development of a deep learning-based Ottoman OCR system presented in [6]. Their model, trained on a hybrid dataset of over 24,000 pages using CNN+BiLSTM architecture, outperformed existing tools such as Google Docs, ABBYY FineReader, Miletos, and Tesseract’s Arabic/Persian models. With character-level recognition rates exceeding 97 after normalization, this model represents a major step forward for printed Ottoman OCR.

Complementing this, the Automated Text Recognition for Ottoman Turkish project from the Digital Ottoman Corpora initiative [7], [8] offers a browser-based HTR system for both printed and handwritten Ottoman texts. It supports collaborative transcription workflows and provides a growing corpus of aligned Ottoman–Modern Turkish pairs. This resource not only contributes to the development of parallel corpora but also presents opportunities to evaluate and train OCR and translation models on real-world data with community engagement.

While these efforts made improvements in Ottoman Turkish translation, parallel corpora and domain-specific models are still limited, which affects effectiveness of machine translation.

B. Datasets for Ottoman Turkish

A key challenge in Ottoman Turkish translation is the lack of parallel datasets. [9] introduced resources such as:

- HisTR (NER dataset with 812 sentences from 17th–19th-century texts).
- OTA-BOUN (Universal Dependencies treebank with 514 annotated sentences).
- Ottoman Text Corpus (OTC) (transliterated texts from the 15th–20th centuries).

Additionally, [1] compiled 13 translated Ottoman Turkish novels, digitized using ABBYY FineReader and ebooklib. To address the corpus scarcity, [10] proposed the DUDU Treebank, a UD-style dataset designed specifically for Ottoman Turkish, ensuring more accurate syntactic and morphological representations.

Further corpus-building efforts include [11], which extracted text from Ottoman periodicals (Sırat-ı Müstakim and Sebilürreşad) to create a cleaner text dataset, further enhancing model pre-training and named entity recognition.

While existing multilingual datasets like OPUS-100 and FLORES-200 could assist in training models, domain-specific parallel corpora remain scarce.

C. MT Models for Low-Resource Languages

Given the limited training data, pre-trained large language models (LLMs) offer promising alternatives. The study in [12] demonstrated that zero-shot and few-shot

prompting improved translation quality for Algerian Arabic to MSA, suggesting similar applications for Ottoman Turkish.

To enhance translation, [13] introduced AlignInstruct which improves low-resource MT through statistical word alignments in BLOOMZ models. Additionally, multi-modal learning, as explored in [14], has shown that image and speech data can enhance translation models, offering future possibilities for historical Ottoman manuscripts.

A comparative study in [15] examined translation methods for another low-resource language, Digaru. It highlighted the benefits of combining SMT for accuracy with NMT for fluency, suggesting that a hybrid approach might be beneficial for Ottoman Turkish translation.

The study in [16] compared different NMT architectures for Turkish, analyzing the impact of sequence-to-sequence models, transformers, and techniques like Byte Pair Encoding (BPE) and Gumbel Softmax. The findings suggest that transformer models with BPE outperform traditional recurrent neural network (RNN) approaches, indicating potential benefits for Ottoman Turkish translation.

III. METHODOLOGY

A. Multimodal Large Language Model (MLLM) Approach

To enable multimodal models to learn the connection between visual and textual elements in *Temettuat* notebooks, a custom dataset was constructed using a layout-aware approach. A page in these notebooks include information about multiple houses which is referred to as "hane" as shown in Fig. 1. A YOLOv5 object detection model was trained to identify and localize specific areas within a given *hane*, such as names, occupations, tax amounts, and household counts, and the model performed very well with identifying different areas in given page. Each identified region was cropped and saved as a separate image file, with filenames indicating the field type. These segmented images were then paired with their corresponding transcriptions and transliterations. The dataset was organized with a JSONL format, where each line represented one *hane* and contained a list of structured region entries, including the image path, the associated ground truth text and the English translation. This structure aimed to simulate a model that would interpret visual input fields and produce corresponding textual outputs.

However, challenges occurred due to the complexity and variability of the handwriting during the experimenting phase with different Multimodal LLM approaches. Several MLLMs were experimented with, including BLIP-2, CLIP, Gemini, DeepSeek-VL, LLaMA.

None of the models generated coherent outputs even though we tried fine-tuning and different configurations. Inconsistencies in layout between images, and the limited availability of high-quality images and the limited size of the dataset made it difficult for the models to learn. As a result, while the multimodal dataset was structurally valid, it lacked the volume and consistency that the models

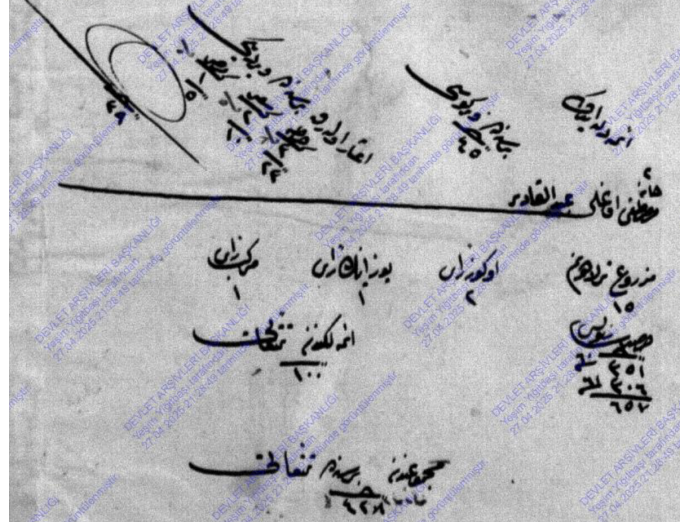


Figure 1. A record of a *hane* in the notebooks.

need to perform effectively on this task. Generated text was often unrelated or repetitive, likely due to domain mismatch and the limitations of current multimodal models in handling low-resource, historical handwritten text.

B. Traditional OCR Based Pipeline

Given the limitations encountered with the multimodal approach, our focus shifted to a more conventional OCR-based pipeline. For this purpose, a separate dataset was prepared at the word level to train a model to recognize individual handwritten Ottoman words and produce their transliterations in Latin characters. Annotations were conducted using Label Studio, where each word on a page was manually marked with a bounding box and labeled with its transliteration. These annotated words were then exported as individual files, with each file containing a single word. Corresponding transliterations were stored in text files sharing the same filename. This format ensured a clear and direct mapping between each image and its textual label. The dataset ultimately contained approximately 980 word-image pairs.

The OCR pipeline was implemented using the TrOCR model from Hugging Face's Transformers library. To prepare the dataset, a custom function loaded all image-text pairs from the folder. For each image file, the script checked for the existence of a corresponding .txt file, and if found, it created a dictionary containing the image path and the label. This resulted in a list of paired samples which were used to create a Hugging Face Dataset object. For preprocessing, light data augmentation techniques, such as rotation and brightness adjustment, were applied to improve model robustness and the TrOCRProcessor was used to convert both images and text into a format suitable for the model. Images were converted into pixel tensors using the processor's image transformation pipeline. Labels (transliterations) were tokenized using

the TrOCR tokenizer and padded or truncated to a maximum length.

This setup enabled efficient batch training using the Hugging Face Trainer API. Despite careful annotation, challenges remained due to the inherent complexity of Ottoman calligraphy and stylistic variations in the handwriting. Although the dataset was relatively small, it offered a more focused and structured input for training OCR models compared to the previous multimodal setup. While TrOCR performed slightly better at correctly identifying couple initial letters and numeric figures the accuracy was not sufficient for practical use in historical research.

IV. EXPERIMENTAL SETUP

A. Hardware

All experiments were conducted using the NVIDIA A100 GPU with high RAM capacity, accessed via Google Colab. This setup provided sufficient computational resources to train and evaluate both multimodal and OCR-based models efficiently.

B. Software & Libraries

The development environment was based on Python 3.10 and utilized the PyTorch framework in conjunction with the Hugging Face Transformers library. The following tools and libraries were employed:

- Label Studio: Used for manual annotation of word-level bounding boxes and transliteration labels.
- YOLOv5: Implemented for object detection to segment specific fields (e.g., names, occupations, taxes) from notebook pages.

All training, evaluation, and data processing were performed within a Google Colab environment, enabling easy access to datasets stored on Google Drive.

C. Datasets

Image Source: Digitized scans of *Temettuat* Notebooks obtained from publicly available Turkish Government archives [17].

Transliterations of the Notebooks: Obtained from the Istanbul Bilgi University Department of History (see Acknowledgements).

Multimodal Dataset: Comprised of 1,288 segmented image regions (e.g., field crops from tax records), each annotated with transliterations and English translations. Structured in JSONL format for layout-aware modeling.

OCR Dataset: Included 980 word-level handwritten image samples, each paired with a manually annotated Latin-script transliteration. The dataset was organized into image-text file pairs for supervised learning.

V. EXPERIMENTAL RESULTS & DISCUSSION

A. Multimodal LLM Approach

The first model we experimented with was BLIP (Boostrapped Language-Image Pretraining). BLIP image captioning model is designed for zero-shot image captioning.

Each cropped image into the model was fed into the model and used its generation capabilities to obtain a textual output from the content. However, BLIP consistently failed to recognize the presence of text in the input images. Since BLIP has no explicit training on images of handwritten or historical text, it could not generalize to the highly localized and script-specific patterns in Ottoman documents.

Next we tried CLIP from OpenAI, a model trained to map images and text into a shared embedding space. Unlike BLIP, CLIP is not generative; it can only measure similarity between an image and a list of text candidates. Since CLIP cannot produce or decode text from the image, so it was ultimately unsuitable for OCR or transliteration tasks.

To test capabilities of a more advanced vision-language model, we incorporated Gemini 1.5 Flash from Google’s Generative AI platform. The model was prompted with each cropped image alongside the several instructions. Its outputs were typically vague or incorrect, and it often produced meaningful words in Arabic letters that does not relate to what is written there. In some cases, it understood the context but it hallucinated when extracting the actual words. In some cases, outputs were descriptive, often referencing paper textures, ink patterns, or interpretations on what kind of text it could be, but none of the responses included actual Ottoman text recognition or meaningful transliteration.

DeepSeek-VL is a vision-language model capable of visual question answering and image captioning with prompting. The model was prompted using cropped images of words and several input queries. Despite its good results on general vision-language benchmarks, DeepSeek-VL was not able to process the Ottoman text meaningfully. This suggests that DeepSeek-VL lacked exposure to Arabic-script handwriting. The model appears to have been trained for descriptive image understanding, not for OCR or low-level script decoding, particularly for non-Latin scripts.

Similarly, we experimented with LLaVA, a model that extends Meta’s LLaMA by integrating a vision encoder and a lightweight projection module to enable image-conditioned generation. Like DeepSeek-VL, LLaVA is designed for prompted tasks involving visual inputs, including captioning, reasoning, and dialogue. I tested LLaVA with similar prompts on the same cropped field images. Again, the model failed to produce any valid transliteration or even recognize that the images contained text while its outputs were often generic.

In summary, despite their state-of-the-art status in general vision-language tasks all tested multimodal models failed to extract usable transliteration data from cropped Ottoman document images. This is largely due to a mismatch between their training domains (natural scenes with English captions) and the target domain (handwritten historical documents in Arabic script). The models

either hallucinated irrelevant visual content or could not represent fine-grained textual information in a way that enables transcription. These results led us to move toward more traditional OCR-based approaches.

B. OCR with TrOCR

The fine-tuning process was carried out using the annotated dataset we created. Despite the limited size of the dataset and the challenge expected by the complex cursive nature of Ottoman handwriting, the model showed slightly better performance than the multimodal LLMs. TrOCR was occasionally able to recognize the initial few characters of a given word correctly and mapping Arabic numbers to their Latin equivalents with high reliability. However, the model still struggled with full-word recognition and often failed to predict more than the first 2–3 letters accurately. These partial predictions indicate that the model was beginning to learn features of the script but lacked sufficient data to generalize effectively across a wide vocabulary.

Nonetheless, this approach was still not sufficient for producing structured and usable transliterations that would be suitable for historians. Achieving higher accuracy with TrOCR would likely require a significantly larger annotated dataset and possibly further domain adaptation. Additionally, training the model on full-line or sentence-level sequences rather than isolated words might improve contextual recognition, although this introduces new challenges related to layout and reading order in historical documents.

In Figure 2 the words "Hasan oğlu Ömer efendi" is written in Ottoman Turkish which translates to "Ömer efendi son of Hasan" in English. Table 1 shows the outputs from each method for a part of the document that is shown in Figure 2.



Figure 2. Hasan oğlu Ömer efendi

VI. CONCLUSION

Although the project did not produced a working model, it provided valuable insight into the challenges of historical OCR and multimodal translation for low-resource, handwritten scripts. Current MLLMs are not well-suited for this kind of structured multimodal input unless they are pre-trained on similar document image data. OCR models like TrOCR offer promise but require significantly more annotated data and possibly domain-specific pretraining to handle Ottoman handwriting.

Multimodal large language models (LLMs) have limitations when used for tasks like recognizing handwritten Ottoman calligraphy. These models are typically trained on natural photos paired with captions, visual question

Table I
OUTPUT OF DIFFERENT METHODS

Model	Output
Actual	Hasan oğlu Ömer efendi
BLIP	a line drawing of a tree branch
Gemini	Here's a breakdown of what's visible: * The Line: This is likely a signature, underlining, or a similar marking. It's not perfectly straight and shows some variations in thickness. * The Background: It appears to be paper with a textured surface. There's faint, illegible text or markings visible underneath, perhaps watermarks or previous writing. * Text to the Right: To the right edge, there's a small portion of dark, illegible text, possibly in Arabic language based on its appearance. The overall impression is that of an aged or worn document where the main focus is the long horizontal line, likely a signature. The quality of the image makes detailed analysis impossible.
Gemini	The visible part of the text seems to be: يعني عزتي امكي (ya'ni) means "that is to say". عزتي (izzati) means "my honor". امكي (amki) possibly from أمكن meaning "to enable".
Deepseek	The overall impression is that of an aged or worn document where the main focus is the long horizontal line, likely a signature. The quality of the image makes detailed analysis impossible.
LlaVa	A close-up image of a faded signature.
TrOCR	Hasy oglu om erty

answering datasets, and some document parsing, which mostly involves printed, English-language texts. As a result, they are not exposed to Arabic-based scripts and are unfamiliar with historical handwriting, such as Ottoman calligraphy. Therefore, they struggle to read or recognize such scripts effectively.

On the OCR side, the challenges are equally substantial. The Ottoman script is highly complex, featuring ligatures, contextual letter forms, ambiguous diacritics, and wide variations in handwriting styles. Moreover, historical documents are often low-resolution or physically degraded, which significantly hampers OCR accuracy. Even advanced models like TrOCR are primarily trained on Latin scripts, and while fine-tuning can help, their architecture and tokenizers are not inherently designed to capture the nuances of Arabic-based scripts, limiting their effectiveness in this context.

VII. FUTURE WORK

In the future work, the dataset can be expanded to enhance the model's learning capabilities. Additionally, exploring semi-supervised learning techniques and synthetic data generation methods for TrOCR could provide significant benefits in improving accuracy and addressing

data scarcity challenges. Hybrid approaches can also be investigated, combining KrakenOCR for raw data extraction with post-processing using LLMs, to refine output quality and enhance overall performance.

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